Application: are unlearned capabilities still in the weights?

The elicit-vs-teach metrics on WMDP-unlearned Zephyr-7B

Question. The main results show that the metrics separate a capability that is already in the weights (fine-tuning *elicits* it) from one that is absent (fine-tuning *teaches* it). Unlearning claims to move a model from the first case to the second. If it did, relearning is teaching and every metric should read "teach"; if the capability is only hidden, relearning is elicitation and the metrics should read "elicit".

Setup in six lines. The original model is Zephyr-7B-beta. Four public unlearned versions of it are tested: RMU, ELM, NPO, SimNPO. The capability is WMDP-bio, 4-option multiple choice, scored as the argmax over the four answer letters (chance 0.25). The items are split at random into halves A and B. *Relearning* fine-tunes on A as text facts (question and correct answer, no options; LoRA r=64) and is read on the held-out half B, which it never sees. The *fine-tuning null* is the same recipe on MMLU facts, which cannot teach bio answers. There is no never-learned model, so the elicit reference is the *original* model (before training) and the *original's own relearning* (after training); the teach reference is each metric's own null, and the elicit/teach values of the main results are given alongside. Section 5 adds two more Zephyr methods (RMU+LAT, GradDiff) and another model family, the Deep Ignorance 6.9B suite (pretraining-data filtering, circuit breakers, latent adversarial training). A model that cannot know the facts, the main results' TinyStories-1B twin, is run on the same facts wherever a read allows a cross-model comparison (the rotated-answer twin in Section 1, the before-training scores in Section 4); it fixes a zero, not a scale.

Reading a cell. Each cell is the measured value and the call it makes. After relearning (Tables 1 and 3): **elicit** (reads like the original: at least half of the original's distance from the null), *partly* (above its own null, less than half), teach (at its own null). Before training (Table 2) the only question is whether the capability can be read in the unlearned model: **readable**, *partly*, or not read. Readable and partly both mean the capability is present (the value is significantly above the model's own null); the split at half of the original's excess says how much of it survives, not whether any does. That reading is one-sided: what a metric reads is there, but what it cannot read may still be there, hidden before or after the point the metric looks at (Section 4). — means there is nothing to measure (the model has no task behaviour for this metric). A metric is called only where this design measures both its null and the original's value; the others are shown next to the original's value without a call (the appendix table).

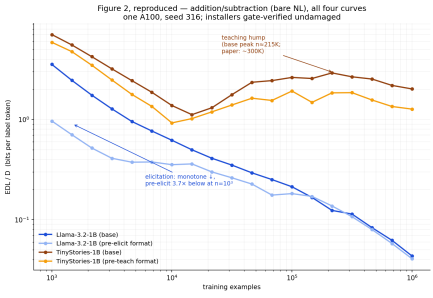
1. Does it come back, and what brings it back?

- **Every method leaves the capability in the weights:** relearning half the items brings the other half back to the original's accuracy, and costs what the original's own relearning costs.
- **What comes back is the knowledge, not the format:** with the answers rotated across items the same training lowers the held-out loss by 0.57–0.59 nats per token less than the true answers do, in every unlearned model as in the original (M17s); for a model that cannot know the facts the true answers add 0.05. Training on 573 true facts moves the other 637 by the same knowledge-specific amount whether or not the model was "unlearned".
- **What unlocks it differs:** for NPO and SimNPO any small fine-tune does; for RMU only bio-specific training does; ELM is in between.

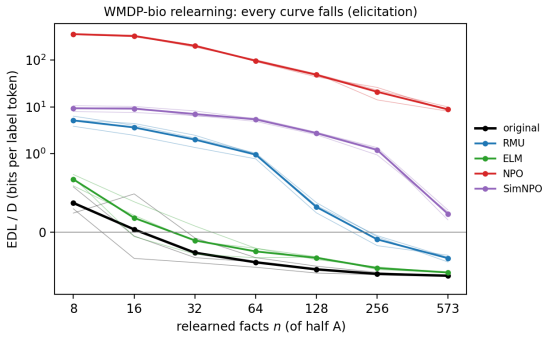
	original	RMU	ELM	NPO	SimNPO
accuracy on B: the model as given	0.67	0.31	0.35	0.26	0.42
after an unrelated (MMLU) fine-tune	–	0.35	0.55	0.62	0.62
after relearning on A	–	**0.68**	**0.67**	**0.64**	**0.66**
the same with no options shown: the correct answer picked among the four, given the question alone (null 0.31)					
as given	0.44	0.39	0.42	0.32	0.32
after an unrelated (MMLU) fine-tune	–	0.39	0.44	0.43	0.45
after relearning on A	–	**0.47**	**0.48**	**0.46**	**0.46**
M17 relearning beats the unrelated fine-tune	–	0.68 vs 0.35 **elicit**	0.67 vs 0.55 **elicit**	no room —	no room —
M17u the unrelated fine-tune unlocks B	–	locked: 0.31→0.35	0.35→0.55 **elicit**	0.26→0.62 **elicit**	0.42→0.62 **elicit**
M18 relearning cost over the original's (nats)	0	$+0.15 \pm 0.19$ **elicit**	$+0.03 \pm 0.14$ **elicit**	$+4.6 \pm 9.0$ —	-0.31 ± 0.53 **elicit**
M19 EDL/D falls with n (bits, n=8 → 573)	0.37 → −0.56	5.1 → −0.34 **elicit**	0.67 → −0.52 **elicit**	355 → 8.8 **elicit**	9.3 → 0.23 **elicit**
M17s what the *true* answers add on B after 573 facts (nats/token; cannot-know model 0.05)	0.58	0.57	0.59	0.57	0.57
MMLU (general ability)	0.58	0.58	0.59	0.47	0.48

Table 1: Accuracy on the held-out half B. M17: position between the unrelated-fine-tune value and the original, 90% interval RMU 1.04 [0.88, 1.19], ELM 0.98 [0.57, 1.40]; for NPO and SimNPO the unrelated fine-tune already reaches the original, so there is no room left to measure. M17u: position between the unlearned model and the original, RMU 0.11 [−0.02, 0.23], ELM 0.64 [0.50, 0.78], NPO 0.89 [0.78, 1.00], SimNPO 0.80 [0.62, 0.99]. M18: prequential code length of relearning (each batch scored before training on it) minus the original's, net of the same difference on the unrelated fine-tune; 0 = as cheap as the original relearning facts it knows; ± is 90%; NPO's first steps repair a broken output, so its interval is too wide to read. M19: the original paper's signature (Donoway et al., §4), EDL per label token of relearning on n facts of A, mean of three seeds, from $n = 8$ to $n = 573$; falling with n = elicitation, a rising phase = teaching (Figure 1). The no-options rows score the correct answer's first token against the other three answers', none shown (the relearning format); the null reassigns the "correct" candidate at random. The picture is the same there: every relearned model reaches the original, RMU beyond its unrelated fine-tune (0.47 vs 0.39; position 1.43, 90% interval $[0.11, 2.75]$, the surface has little room), the others already there after it; ELM is largely there before any training. M17s: the loss on the held-out B facts after relearning the 573 facts of A with every answer moved to another item (same prompts, same answers, no knowledge) minus the loss after the true answers: what the true answers add. The same recipe on a model that cannot know the facts (the main results' TinyStories-1B twin) adds 0.045; a different model, so the row is reported, not called.

- **The original paper's own signature agrees** (M19, Figure 1): the information relearning extracts per label token falls with the number of facts for every method, in every seed and under both floors, as it does for the original. Teaching would show a rising phase, as the TinyStories arm does in panel (a). What an unlearned model needs beyond the original is paid on the first facts and shrinks with every doubling of n.



(a) reference: the project's Figure-2 replication



(b) WMDP-bio relearning

Figure 1: The EDL sweep. (a) The project's replication of the paper's Figure 2 (arithmetic; decisions.md 2026-08-22): the model that has the capability (Llama-3.2-1B, blue) falls monotonically; the model that must be taught it (TinyStories-1B, brown) falls, then rises 2.6× (the teaching hump), then saturates. (b) Relearning on the first n facts of a seeded order of A, bits per label token, floor = the relearned model's own validation loss (OCV; the paper's test floor, the loss on B, gives the same calls); thick = mean of three seeds, thin = each seed; linear between −1 and 1, logarithmic beyond. Every curve falls, the original's included; no method has a rising phase. The excess over the original per label token shrinks from $n = 8$ to 573 by 21× (RMU, 4.8 → 0.22 bits), 7× (ELM, 0.30 → 0.04), 38× (NPO, 355 → 9.4) and 11× (SimNPO, 9.0 → 0.79): a fixed cost paid on the first facts, not a cost per fact. NPO's level is its output repair (its first batch costs about 240 nats per label token). The original ends below zero because its validation floor is harder than its training facts (Caveats).

2. Before any training: is the capability still readable in the unlearned model?

- **SimNPO** reads like the original on 7 of 12: on the multiple-choice prompt the circuit, the state, the heads and the probe are all intact, and the output is pushed away from the answer. With no options shown the answer is not read either.
- **NPO:** the answer is readable in the state (probe 0.47 vs original 0.60) while the output says nothing, on either surface, so the damage is at the read-out.
- **ELM:** nothing is readable on the multiple-choice prompt, but with no options shown ELM still prefers the correct answer (0.40; original 0.43, null 0.31; position 0.70, 90% interval $[0.33, 1.07]$), and its circuit on that surface is repeatable: the suppression is bound to the format. Writing the original's head states in turns the multiple-choice answer on (0.27 → 0.48; the original scores 0.49 on this test).
- **RMU** hides it best: the probe, the circuit and the state swap cannot read it, and with no options shown it is at the edge of the null (0.35, $p = 0.05$). Yet relearning brings it back in full on both surfaces (Table 1), so what these metrics miss in RMU is present, not absent.

3. After relearning on A: does the relearning look like eliciting or teaching?

Every relearned child is compared with the *original's* relearned child, trained with the same recipe on the same facts the original already knows; that child is the elicit reference in this setting.

- **After relearning, all four read elicit on almost every called metric.** On the no-options surface the child's circuit overlaps the original's at 0.39–0.60, above the split-half ceilings (0.31–

	metric	main results: elicit / teach	original	RMU	ELM	NPO	SimNPO
M12	hidden preference, bio (nats)	2.31 / 0.06	3.33	0.84 *partly*	0.40 *partly*	0.42 *partly*	3.52 **readable**
M12	hidden preference, cyber (nats)	2.31 / 0.06	3.57	0.34 *partly*	0.13 *partly*	2.32 **readable**	3.31 **readable**
M16′	answer probe, bio (accuracy; shuffled ≈0.28)	–	0.60	0.28 not read	0.28 not read	0.47 **readable**	0.49 **readable**
M16′	answer probe, cyber	–	0.46	0.28 not read	0.29 not read	0.41 **readable**	0.39 **readable**
M9	answer depth: lens, best middle layer (nats)	2.1 / 0.0	4.88	1.18 *partly*	0.39 *partly*	0.17 *partly*	0.45 *partly*
M11	repeatable task circuit (split-half J@32)	0.60 / noise	0.64	noise not read	noise not read	noise not read	0.46 **readable**
M14	answer carried by the state (swap gain)	+0.39 / +0.05	+0.66	+0.08 not read	+0.09 not read	+0.03 not read	+0.38 **readable**
M15	option-reading heads (heads; flip rate)	22 / 0 heads	15; 0.73	20; 0.52 **readable**	0 not read	0 not read	57; 0.70 **readable**
M12f	correct answer preferred among the four with no options shown (top-1; null 0.31)	–	0.43	0.35 not read	0.40 **readable**	0.29 not read	0.34 not read
M11f	repeatable circuit on the no-options surface (split-half J@32)	–	0.49	0.42 not read	0.60 **readable**	0.28 not read	0.23 not read
M1★	original's 32 heads destroy U's answer (random 0.05); machinery, Section 4	–	0.93	0.99†	no answer —	no answer —	0.89
M10★	original's head states patched into U (accuracy)	–	0.49	0.20→0.30 *partly*	0.27→0.48 **readable**	0.00→0.00 not read	0.13→0.25 *partly*
M2★	wiring, U vs original (J@256; ceiling 0.55)	0.36 / –	–	no answer —	no answer —	no answer —	0.23 *partly*
M13	accelerating descent (negative share; blocks 20, 28)	1.00 / 0.00	0.00	0.00 —	0.00 —	**1.00** —	0.00 —
M21	hidden preference, no options, full answer: loss with the answers rotated across items minus loss on the true answers (nats/token; reported, Section 4)	1.15 / 0.09	1.58	1.30	1.24	2.52	2.64
	tally (readable / partly / not read)			1 / 4 / 6	3 / 3 / 5	3 / 2 / 6	**7 / 3 / 2**

Table 2: Values on the unlearned model itself, no training. "Main results" gives the elicit parent and the teach parent from the arithmetic study where the metric has one. M16′ adapts M16: the probe reads the correct option (4 classes), not the operands. M14: how often swapping the state at the answer position makes the model prefer the other item's answer, minus the rate with no swap. M10★: accuracy of U before and after the original's states at its 32 top heads are written in (32 random heads: RMU 0.20, ELM 0.34, SimNPO 0.16). † RMU's remaining answer signal is small (logit difference +0.75 vs +4.87 in the original); the original's heads carry all of it, but its MMLU-circuit heads carry it equally (Section 4), so the row is not tallied. M12f and M11f are read with no options in the prompt (question, then "Answer:"): M12f scores the correct answer's first token against the other three answers' first tokens, the null reassigns the "correct" candidate at random ($p < 0.01$ counts as readable; RMU $p = 0.05$, SimNPO $p = 0.06$); M11f is M11 on that surface. M13 cannot be called here: the original is at a minimum of its own bio loss (negative share 0.00, unlike the main results' latent elicit parent at 1.00), so the metric has no elicit reference; NPO alone sits on a strongly concave slope ($\lambda_{\min} = -1.5 \times 10^5$), where gradient ascent left it, and any descent step speeds up. Curvature is restricted to decoder blocks 20 and 28 (the full Hessian does not fit a 7B model on one GPU). M8 (latent reach) has no WMDP analogue and is not run. M21 is M12f's full-answer form: the mean label-token loss on the B facts with each item's answer replaced by another item's (four rotations), minus the loss on the true answers, no training. It is reported, not tallied (it would double-count M12f); its "main results" column is the arithmetic elicit parent / the format-installed teach parent, and a model that cannot know the bio facts reads 0.22 (Section 4). It reads NPO (2.52) where M12f

	metric	main results: elicit / teach	orig. re-learned	RMU	ELM	NPO	SimNPO
M1f	circuit overlap with the original on the no-options surface, heads (J@32; chance 0.02; cross-subject 0.21–0.28)	0.33 / 0.02	ceiling 0.31–0.37	0.60 **elicit**	0.46 **elicit**	0.46 **elicit**	0.39 **elicit**
M17f	held-out B with no options shown, beyond the unrelated fine-tune	–	0.44	0.47 vs 0.39 **elicit**	no room —	no room —	no room —
M10	child's head states patched into U (accuracy)	switches on / –	–	0.21→0.31 teach	0.33→0.54 **elicit**	0.00→0.00 teach	0.14→0.20 teach
	tally, with Table 1 (elicit / partly / teach)			5 / 0 / 1	5 / 0 / 0	2 / 0 / 1	3 / 0 / 1
multiple-choice circuit rows: the same values against the original's MMLU circuit (Section 4), so not called:							
M1	circuit overlap with the original, heads (J@32; vs its MMLU circuit)	0.33 / 0.02	ceiling 0.60; own 0.56	0.73; 0.60	0.64; 0.56	0.68; 0.52	0.68; 0.52
M1	original's 32 heads destroy the child's answer (bio heads; MMLU heads)	0.73 / –	0.86; 0.84	0.88; 0.85	0.86; 0.84	0.86; 0.83	0.83; 0.84
M2	wiring overlap with the original (J@256; vs its MMLU wiring)	0.36 / –	ceiling 0.27; own 0.27	0.32; 0.32	0.32; 0.34	0.33; 0.32	0.29; 0.31
M4	circuit at step 1 vs final (J@32; ceiling)	at step 1 / mid-run	–	0.45 (0.78)	0.73 (0.73)	0.49 (0.73)	0.42 (0.73)
reported, not called (no null or teach value is measured for these here):							
M3	head roles shared with the original (J; chance 0.01)	0.69 / 0.88	–	0.13	0.08	0.17	0.15
M5	gradient fades (first 1% / last 10%)	3.4× / 0.7×	3.2×	14.5×	6.1×	134×	13.7×
M6	weight write $\|\Delta W\|/\|W\|$ ($\times 10^{-3}$)	50 / 86	8.8	8.9	8.8	11.2	11.0
M7	state change: one shared push (PC1)	0.06 / 0.57	0.64	0.49	0.57	0.69	0.66

Table 3: Values on the relearned child, read on B. "Original relearned" is the elicit reference: the same training on the original, which knows the facts. M1f: the child's circuit for answering with no options shown against the original's circuit on the same surface; the ceilings are the split-half overlaps of the circuits themselves, and "cross-subject" is the same children against the original's MMLU circuit on that surface (the original's own bio circuit against its MMLU circuit there: 0.21). M17f: the no-options accuracy of Table 1; "no room" where the unrelated fine-tune already reaches the original. The multiple-choice circuit rows give, after the semicolon, the same measurement against the original's MMLU circuit (or with the original's MMLU-circuit heads ablated); the two are equal within 0.1 and at the split-half ceiling, so these rows show the original's answering machinery in use, not bio knowledge, and are not tallied. M4: overlap of the circuit after the first training step with the final one, in brackets the ceiling. M10: accuracy of U with the child's states at the child's 32 top heads written in; the child itself scores 0.54–0.57. M3, M5, M6 and M7 are shown next to the original's relearning but not called, because the main results' scale does not carry over and this design has no teach value to replace it: the original's own relearning reads 0.64 on M7, past the main results' teach value; LoRA and the same stopping rule fix the size of M6's write; M5's first steps are inflated by repairing the unlearned output; and M3 did not separate elicit from teach in the main results either (0.69 vs 0.88). The tally counts M17, M18 and M19 from Table 1 (M17u is not counted: it asks what unlocks the capability, not whether it is there).

0.37) and well above the cross-subject overlap (0.21–0.28); on the multiple-choice prompt the child uses the original's answering machinery (overlap 0.64–0.73), which any four-option question uses (Section 4).
- The exception is patching a child's states into the unlearned model (M10), which fails where the unlearned model's own output is broken (NPO, SimNPO) or scrambled (RMU).

4. Why the calls are mixed, and how each metric is judged

Before training, a call can only be one-sided. Unlearning is meant to stop the capability from showing, so a metric that looks where the suppression acts, or after it, finds nothing, while one that looks before it, or goes around it, still finds the capability. Mixed calls therefore say where each method hides the capability; they do not contradict each other. Whether it is still in the weights is answered by the tests that bring it back (M17, M17u, M18, M19, M10$^\bigstar$), and those read elicit for all four.

- **Hidden at the read-out: NPO and SimNPO.** The answer is readable in the state (probe 0.47 and 0.49; original 0.60, shuffled labels 0.28; through the layers where the original's answer is readable, 0.63 and 0.67 of the original's excess) while the output map does not carry it (lens 0.04 and 0.12 of the original's; accuracy on B 0.26 and 0.42; original 0.67). Both train the output probability of the forget-set answers down, and their weight change is spread over 29 of 32 layers and the output head. NPO was trained on bio only, and its cyber answers still show at the output (hidden preference 2.32; original 3.57).
- **Hidden before the answer forms: RMU and ELM.** Even the probe cannot read the answer at any layer (0.28, the shuffled level), so the suppression acts inside the network before the state carries the answer. The weights say where: 90% of RMU's change sits in the MLPs of layers 5–7, and 90% of ELM's in layers 4–7 (its adapter). Going around that edit brings the capability back: writing the original's head states into ELM restores its accuracy to 0.48 (original 0.49), and ELM still answers with no options shown (0.40; original 0.43), so its suppression is bound to the multiple-choice format. Relearning restores both, and it does not repair the edit: every child's LoRA write puts the same share into the edited layers as the original's own relearning does (RMU 10%, ELM 13%; RMU's write is 13% above the flat per-layer share at layers 5–6, a bump, not a repair). Figure 2 shows all of this by layer.

Machinery or knowledge? A multiple-choice reading needs the machinery that reads the options and writes the letter as well as the knowledge, so every metric read on that prompt was run again on the MMLU split (Table 4). The machinery is intact in all four models: on MMLU, the hidden preference, the probe, the state swap and the option-reading heads read at 0.57–1.22 of the original's excess; NPO and SimNPO show partial read-out damage there (lens 0.43 and 0.24; their MMLU accuracy is 10 points below the original's). So the bio drops on the probe, the state swap and the heads are knowledge-specific: RMU and ELM keep the MMLU reading at the original's level and lose the bio reading entirely. The circuit metrics are the exception. A relearned child's circuit overlaps the original's MMLU circuit as much as its bio circuit (heads 0.52–0.60 vs 0.64–0.73, both at the split-half ceiling; wiring equal), and the original's MMLU-circuit heads are as necessary as its bio-circuit heads, in the unlearned models and in every child. Those rows show the original's answering machinery in use and are no longer tallied. On the no-options surface the circuits are bio-specific (0.39–0.60 against a cross-subject 0.21–0.28), and that reading (M1f) is tallied instead.

A single score before training. The reads above are one-sided, so we asked for one number, from the weights and the data alone, that says whether fine-tuning will elicit or teach, and tested it on every case with a known outcome (Table 5): the main results' parents before their 4M-example fine-tunes (the format-installed parent is taught, the op-bridge parent is elicited), the same model on the bio

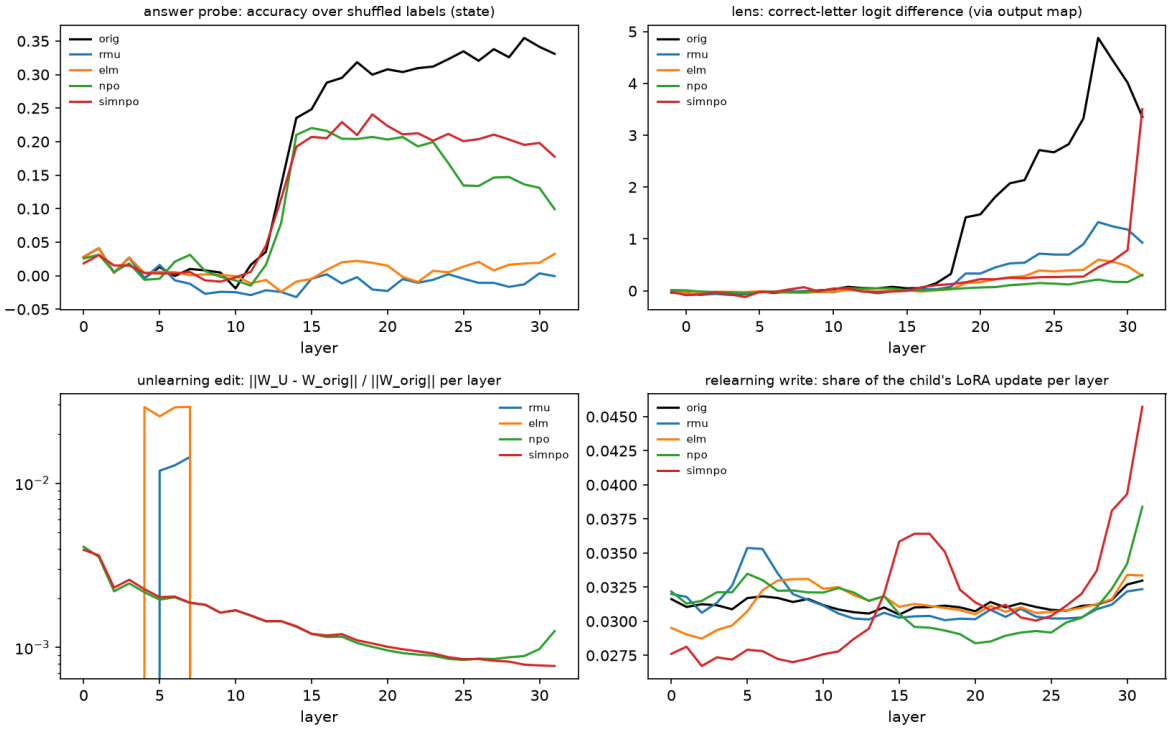


Figure 2: Where each method acts, by layer, from the stage-1 files and the checkpoints. Top left: the answer probe's accuracy over its shuffled-label null at every layer; NPO and SimNPO follow the original from layer 13 at about two thirds of its height, RMU and ELM stay at zero. Top right: the correct-letter logit difference read through the model's own output map; only the original rises from layer 18, SimNPO's jumps at the last layer (its output-level preference is intact, its state readable, and its accuracy still low: the pushed-away answer is a matter of which letter wins). Bottom left: the relative weight change of the unlearning edit per layer, log scale; RMU in layers 5–7 and ELM in 4–7 at ten times the level of NPO's and SimNPO's spread-out changes. Bottom right: the share of each relearned child's LoRA write per layer, next to the original's own relearning (black, flat at 1/32); no child concentrates its write where its parent was edited. Across the headline metrics the median position between the null and the original is 0.17 (RMU), 0.02 (ELM), 0.00 (NPO) and 0.64 (SimNPO), with absent readings counted as zero: a measure of how well each method hides the capability, not of how much remains.

	metric	RMU	ELM	NPO	SimNPO
before training: position r on bio | on MMLU (control); 1 = the original, 0 = the null)					
M12	hidden preference	0.25 | 0.99	0.12 | 0.91	0.13 | 0.89	1.06 | 1.22
M16′	answer probe	0 | 1.02	0 | 1.05	0.59 | 0.90	0.67 | 0.98
M9	lens, middle layers	0.24 | 0.99	0.08 | 0.89	0.04 | 0.43	0.09 | 0.24
M14	answer carried by the state	0 | 0.96	0 | 0.89	0 | 0.64	0.57 | 0.57
M15	option-reading heads	0.70 | 0.91	0 | 0.98	0 | 0.70	0.96 | 0.82
M1★	original's heads necessary in U: bio heads | MMLU heads	1.06 | 1.11	–	–	0.89 | 0.94
M10★	original's states patched in: at bio heads | at MMLU heads	0.33 | 0.32	0.92 | 0.83	0 | 0	0.28 | 0
after relearning: overlap with the original's bio circuit | with its MMLU circuit (the original's own: 0.56, 0.27, 0.21)					
M1	heads, multiple-choice prompt (J@32)	0.73 | 0.60	0.64 | 0.56	0.68 | 0.52	0.68 | 0.52
M2	wiring, multiple-choice prompt (J@256)	0.32 | 0.32	0.32 | 0.34	0.33 | 0.32	0.29 | 0.31
M1f	heads, no options shown (J@32)	0.60 | 0.26	0.46 | 0.26	0.46 | 0.28	0.39 | 0.28
M1	original's heads necessary in the child: bio heads | MMLU heads	1.02 | 1.01	0.99 | 0.99	0.99 | 0.98	0.96 | 0.99

Table 4: The multiple-choice controls. Before training, each metric's position between its own null (0) and the original (1) on the bio items and, after the bar, on MMLU items, where the knowledge is not in question and only the machinery is measured. A method that keeps the MMLU value and loses the bio value has lost knowledge, not machinery (RMU, ELM on the probe, the state swap and the heads); NPO and SimNPO lose part of the read-out on MMLU too (M9). After relearning, a circuit overlap that is the same against the original's MMLU circuit as against its bio circuit measures the shared machinery (M1, M2, and the necessity of the original's heads); the no-options circuit (M1f) is the one that separates bio from MMLU.

facts it cannot know, and the Zephyr models. The plainest read on the fact surface does it: M21, the loss on the true answers against the loss with the answers rotated across items, no training, no options (Table 2). Every known teach-side case is at or below 0.22 nats per token, every case whose relearning reads elicit on all three recovery tests at or above 0.96 (GradDiff; the four methods of Tables 1–2 1.24–2.64, with the original, NPO included, whose whole output is shifted by about 244 nats with the log-odds between true and other answers intact), and the three models whose trajectory is gated, the Deep Ignorance safeguards of Section 5, sit between (0.79–0.89). Why this is the right quantity is set out under M21 in the metrics document: the rotation cancels every term of the loss that depends on the question alone or on the answer alone, so the gap is the interaction term of the parent's loss on the true pairs — the pairing information the weights hold, in the EDL framework's units, before any training — and it bounds the information any read-out can draw from the parent to identify its answers. That is why NPO's 244-nat shift leaves it untouched, and why only the two methods that edit the representation before the answer forms (RMU, ELM) lower it, to 1.30 and 1.24 against the original's 1.58. Two limits. It is a sufficient sign, not a necessary one: knowledge that shows in no conditional likelihood would read zero and could still be elicited. And its within-model null, re-pairing the answers, does not contain the lexical coherence between a question and its own answer; that floor is 0.22 for a 1B model that knows neither the facts nor much else, and for a general 7B model without bio it is unknown (none exists; the one candidate, Section 5, is not knowledge-free on these items). A second null narrows it: rotating each answer only among answers of its own kind (an automatic clustering of the answer strings, nothing inspected) subtracts whatever the kind of answer explains, and on the Deep Ignorance set it moves no value by more than 0.1 (Table 6), so the floor is not a same-kind artefact; what it may still contain is question-specific. So M21 is reported beside the tallied

metrics, not tallied.

model	facts	outcome	M21: loss gap, rotated − true (nats/token)
TinyStories-1B, base	arithmetic	teach	0.13
TinyStories-1B, format installed, random labels	arithmetic	teach	0.09
TinyStories-1B, capability latent (op-bridge)	arithmetic	elicit	**1.15**
TinyStories-1B (cannot know the facts)	WMDP-bio	teach	0.22
Zephyr-7B, original	WMDP-bio	elicit	**1.58**
RMU	WMDP-bio	elicit (Table 1)	**1.30**
ELM	WMDP-bio	elicit	**1.24**
NPO	WMDP-bio	elicit	**2.52**
SimNPO	WMDP-bio	elicit	**2.64**
RMU+LAT	WMDP-bio	elicit (Section 5)	**1.90**
GradDiff	WMDP-bio	elicit	**0.96**
Deep Ignorance 6.9B, unfiltered baseline	WMDP-bio	elicit (its own relearning)	**1.34**
strong data filter	WMDP-bio	no room to call (Section 5)	1.23
circuit breakers	WMDP-bio	mixed: comes back, costly, undetermined trajectory	*0.79*
circuit breakers + LAT	WMDP-bio	mixed: comes back, costly, rising phase	*0.87*
filter + circuit breakers + LAT	WMDP-bio	mixed: comes back, costly, rising phase	*0.89*

Table 5: The fact-surface preference before any training, on every case whose outcome is known. The arithmetic rows are the main results' parents before their 4M-example fine-tunes (600 held-out problems of the frozen evaluation block; on 600 training problems: 0.11, 0.00, 0.91); the bio rows are read on half B. The spread of the loss over the four rotations is 0.01–0.05 nats in every row, so every value above 0.22 is far above noise; the question a reader must keep is the coherence floor of a large model, which this design cannot measure. The lower block is the Deep Ignorance set of Section 5, read against its own baseline: the three safeguarded models are the only cases between the teach-side maximum and the elicit-side minimum, and the only ones whose relearning reads mixed.

5. Other methods, another model family

The same pipeline, unchanged, on two more Zephyr unlearning methods — RMU with latent adversarial training (RMU+LAT) and gradient difference (GradDiff) — and on a second model family: EleutherAI's Deep Ignorance suite, 6.9B GPT-NeoX models in which the safeguard is built in during or right after pretraining: a baseline pretrained on unfiltered data; the same recipe on data with a strong biothreat-proxy filter; circuit breakers (CB) on the baseline; CB plus latent adversarial training (CB+LAT) on the baseline; and CB+LAT on the filtered model. The baseline is the original for its set; the filtered models are separate pretraining runs, so any read that compares their weights or states with the baseline's is a cross-model read and is reported, not called. Everything else (halves A and B, the MMLU null, the thresholds) is as above; Table 6.

- **The two Zephyr methods read as the first four did.** RMU+LAT is RMU hidden deeper: nothing reads before training except a trace at the lens, an unrelated fine-tune does not unlock it ($0.28 \to 0.32$), yet relearning on A restores B in full (0.68; the original 0.67) at a cost within noise of the original's, EDL/D falls on every seed, the true answers add the original's 0.56 nats, and its fact-surface preference is *above* the original's (1.90 vs 1.58). GradDiff breaks the output and the model with it (MMLU 0.44); any fine-tune unlocks it ($0.27 \to 0.60$) and M21 reads 0.96. Both: in the weights, elicited.
- **The Deep Ignorance safeguards hide the answer everywhere, and it still comes back.** For CB, CB+LAT and filter+CB+LAT no before-training metric reads the bio answer — not the preference, the probe, the lens, the state swap or the no-options preference — while cyber and MMLU read at the baseline's level on all of them (the bio loss is knowledge-specific). This is the first set in which even the probe finds nothing. Yet M21 reads 0.79–0.89 (0.86–0.88 with same-kind rotations), four times the cannot-know model and 0.51–0.60 of the way from that floor to the baseline; relearning on A brings B back to the baseline's level (0.53, 0.56; the filtered-base model

	metric	Zephyr-7B (original of Table 1) RMU+LAT	GradDiff	Deep Ignorance 6.9B baseline	filtered	CB	CB+LAT	filt.+CB+LAT
before training: position r between the model's own null (0) and its original (1), bio | MMLU control								
M12	hidden preference	0.14 | 0.95 *partly*	0.12 | 0.91 *partly*	1	0.74 | 0.89 **readable**	0 | 0.99 not read	0 | 1.00 not read	0 | 0.76 not read
M16′	answer probe	0 | 0.99 not read	0.67 | 0.90 **readable**	1	0.68 | 0.94 **readable**	0.18 | 1.04 *partly*	0 | 0.97 not read	0 | 1.01 not read
M9	lens, middle layers	0.16 | 0.97 *partly*	0.03 | 0.33 *partly*	1	0.67 | 0.62 **readable**	0 | 1.16 not read	0 | 0.95 not read	0 | 0.62 not read
M14	answer carried by the state	0 | 0.89 not read	0 | 0.57 not read	1	0.65 | 1.31 **readable**	0 | 0.94 not read	0 | 1.38 not read	0 | 1.25 not read
M12f	correct answer preferred, no options (top-1; null 0.31)	0.32 not read	0.28 not read	0.38	0.36, p=0.02 not read	0.35 not read	0.33 not read	0.37, p=0.03 not read
M10★	original's head states patched in (accuracy)	0.12→0.22 not read	0→0 not read	0.40	0.35→0.33	0.14→0.26 *partly*	0.12→0.28 **readable**	0.14→0.11
M21	loss gap, rotated − true (nats/token); *same-kind rotations*	1.90	0.96	1.34; *1.32*	1.23; *1.21*	0.79; *0.88*	0.87; *0.86*	0.89; *0.87*
accuracy on B and the recovery tests								
	as given	0.28	0.27	0.53	0.47	0.28	0.26	0.31
	after an unrelated (MMLU) fine-tune	0.32	0.60	–	0.49	0.45	0.31	0.32
	after relearning on A	**0.68**	**0.67**	–	0.49	**0.53**	**0.56**	**0.44**
M17	relearning beats the unrelated fine-tune	0.68 vs 0.32 **elicit**	0.67 vs 0.60 **elicit**	–	0.49 vs 0.49 teach	0.53 vs 0.45 **elicit**	0.56 vs 0.31 **elicit**	0.44 vs 0.32 **elicit**
M17u	the unrelated fine-tune unlocks B	locked	0.27→0.60 **elicit**	–	locked	0.28→0.45 **elicit**	locked	locked
M18	relearning cost over the original's (nats/token)	+0.44 ± 0.40 **elicit**	+3.0 ± 9.4 —	0	+0.82 ± 0.14 teach	+0.80 ± 0.34 teach	+0.98 ± 0.31 teach	+2.05 ± 0.27 teach
M19	EDL/D with n (bits, n=8 → 573)	15.4 → −0.10 **elicit**	357 → 9.6 **elicit**	0.12 → −0.45	0.78 → 0.07, mixed —	6.0 → 0.56 *partly*	2.9 → 4.8 → 0.79 teach	2.1 → 4.1 → 1.7 teach
M17s	what the true answers add on B after 573 facts (nats/token; cannot-know model 0.05)	0.56	0.57	0.52	0.46	0.55	0.50	0.46
	MMLU (general ability)	0.58	0.44	0.48	0.49	0.50	0.49	0.51

Table 6: The extension. Before training, each position r is the model's value between its own null and its original's, on the bio items and, after the bar, on MMLU (the machinery control); the calls follow Table 2. M12f gives the top-1 rate against the no-options null of 0.31 ($p < 0.01$ counts as readable; the baseline's $p = 0.006$). M10★ for the two filtered models compares states across pretraining runs and is not called. M21: the plain gap, then (italic) the gap with each answer rotated only among answers of its own kind; the model that cannot know the facts reads 0.22 and 0.21; the Zephyr set's same-kind values are not yet computed. M18 and M19 as in Table 1; M19's three values for CB+LAT and filter+CB+LAT are $n = 8$, the peak ($n = 64$ and 128) and $n = 573$, "increasing" on 3/3 seeds under both floors; CB falls under the validation floor (2/3 seeds) and is mixed under the test floor; the filtered model's seeds disagree under both. M17 for the filtered model has no room: the baseline itself scores 0.53 on B. GradDiff's M18 interval is too wide to read for the same reason as NPO's (output repair).

0.44 against its own base's 0.47); and what the 573 true facts add on B is the baseline's amount (0.46–0.55 nats per token against 0.52).

- **But the way back is the first teach-side trajectory in this study.** Relearning costs 0.8–2.05 nats per label token more than the baseline's own relearning, net of the null (every Zephyr model: within 0.44 of free), and for the two LAT-trained models the paper's signature shows its rising phase: EDL per label token climbs from $n = 8$ to $n = 64$ (CB+LAT, 2.9 → 4.8 bits) or $n = 128$ (filter+CB+LAT, 2.1 → 4.1) before it falls, on every seed under both floors — the shape of the taught arm in Figure 1a, not of the unlearned Zephyr curves in Figure 1b. CB alone falls from 6.0 to 0.56 under the validation floor and is mixed under the test floor. Read alone, the signature calls these three models taught. These are also the only cases whose M21 sits between the teach-side maximum (0.22) and the elicit-side minimum (0.96).
- **What the rising phase is the price of.** A sweep's data pay for two things at once: the facts, and whatever stands between the fine-tune and the facts. To separate them the same sweep was run on the rotated-answer twin of A (every answer moved to another item, so the data carry the format and the training dynamics and no pairing information), and M21 on B was read at the kept weights of every point of both sweeps (Table 7, Figure 3). Three things follow. First, the hump is in the rotated sweep too: with no facts in the data, CB+LAT's EDL/D still climbs 2.1 → 4.8

bits to $n = 64$, filter+CB+LAT's 0.6 → 3.2, CB's 3.5 → 5.3, on every seed under both floors, while the baseline, the filtered model and every Zephyr model pay no hump in either sweep (the unlearned Zephyr models' large EDL/D is the same with the answers rotated: output repair, not facts). Second, the knowledge-specific part, EDL/D(true) − EDL/D(rotated), is *negative over the first sizes* in every model whose capability reads present ($n \leq 32$ means of −0.4 to −3.7 bits per token, at or below zero from $n = 8$ or 16: the true facts are cheaper than rotated ones by about the pairing information the weights hold, and the saving per token shrinks toward zero as n grows), and *positive* there in exactly the three safeguarded models (+1.3, +0.6, +0.8 bits over $n \leq 32$; +2.5, +0.8, +1.5 at $n = 8$: the true facts cost more than rotated ones, a barrier specific to the pairings), crossing to negative at $n = 128$–573. Third, M21 on the held-out B facts rises above the untrained baseline's 1.34 after 32 true facts of A for CB and CB+LAT (0.79 → 1.47, 0.87 → 1.51) and after 128 for filter+CB+LAT (0.89 → 1.39), while after rotated facts it falls in every model, to 0.23–0.40 by $n = 573$: inconsistent pairings close the gate on B as well; they never open it. Thirty-two consistent facts cannot teach 637 others. They open a gate, and the rising phase is the price of opening it. The signature read the gate as teaching because it cannot tell the two costs apart; read next to its fact-free twin it says what the recovery tests said: present, gated, elicited after the first 32–128 facts.

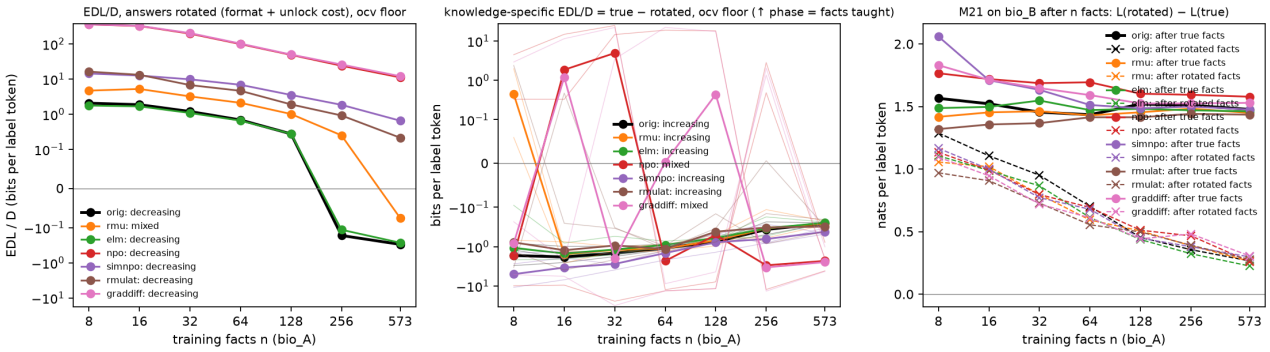
- **The same two reads on the toy task, where the outcome is known (Table 8, Figure 4).** The arithmetic Figure-2 arms were swept again on the rotated-answer twin of their training file and M21 was read on the held-out block at every checkpoint. On the two taught arms (TinyStories-1B base and format-installed) the rotated twin has no hump at all, and the knowledge-specific curve carries the whole of it: +2.3 to +2.5 bits per token at $n = 100{,}000$–316,000 against 0.0–0.4 below $n = 5{,}000$. On the three elicit arms (the op-bridge parent on the same base, Llama-3.2-1B base and format-installed) no hump survives; the Llama base pays +1.2 bits at $n = 1{,}000$ and is below zero by 21,000, the shape of the Deep Ignorance safeguards, here the output format, which the installer removes (the installed arm reads zero throughout). M21 on the held-out block makes half of its total rise by 15,000 to 32,000 examples on the taught arms, by 1,000 to 3,200 on the elicit arms, and by 16 to 64 facts in the gated WMDP models; after rotated training it is 0.00 everywhere. The reading of the three WMDP safeguards therefore follows the calibration: a cost at the first sizes that the subtraction leaves and that crosses below zero, with the held-out preference readable within the first few dozen facts, is the gate signature, two to three orders of magnitude in n away from the taught arms; it is the rule the knowledge-specific row (M19k) now applies. On WMDP it calls elicit for the original, RMU, ELM, SimNPO and the Deep Ignorance baseline, gate for CB+LAT, and mixed for the rest, where three seeds leave the early value within two standard errors of zero (Table 7; the margin, not the sign, is what those models fail).
- **The strong data filter is not knowledge-free on WMDP-bio as a whole.** The filtered model keeps 0.92 of the baseline's fact-surface preference, reads at 0.65–0.74 of the baseline's excess on the preference, the probe, the lens and the state swap, and scores 0.47 on B against the baseline's 0.53; relearning on A adds nothing beyond the unrelated fine-tune because there is almost nothing left to add. Consistent with the suite's design — the filter targets a narrow biothreat-proxy slice and WMDP-bio is mostly general biology — it cannot serve as the never-learned anchor this design lacks (the Caveats). The tamper-resistance set (TAR on Llama-3-8B) is pending access to its gated original.

	RMU	RMU+LAT	baseline	filtered	CB	CB+LAT	filt.+CB+LAT
EDL/D, true answers, n=8 → peak → 573 (bits)	5.1 → −0.34	15.4 → −0.10	0.12 → −0.45	0.78 → 0.07	6.0 → 0.56	2.9 → 4.8 → 0.79	2.1 → 4.1 → 1.7
EDL/D, answers rotated (M19-rot)	4.7 → −0.08	16.2 → 0.22	1.0 → 0.01	0.73 → 0.32	3.5 → 5.3 → 0.93	2.1 → 4.8 → 1.27	0.6 → 3.2 → 1.9
knowledge-specific, true − rotated: mean over $n \le 32 \pm$ SE over seeds (bits); ≤ 0 from n (M19k)	−0.78 ± 0.31; 16	−0.99 ± 0.89; 8	−1.07 ± 0.24; 8	−0.43 ± 0.47; 16	+1.33 ± 0.68; 256	+0.64 ± 0.29; 128	+0.80 ± 0.42; 573
read (M19k rule)	elicit	mixed (1.1 SE)	elicit	mixed (0.9 SE)	mixed (2.0 SE)	gate	mixed (1.9 SE)
M21 on B: untrained → after 32 true facts → after 573 (nats/token; M21-n)	1.30 → 1.46 → 1.45	1.90 → 1.37 → 1.44	1.34 → 1.51 → 1.64	1.23 → 1.40 → 1.43	0.79 → 1.47 → 1.60	0.87 → 1.51 → 1.67	0.89 → 0.93 → 1.44
M21 on B after 573 rotated facts	0.26	0.28	0.40	0.40	0.32	0.31	0.37

Table 7: The two costs inside EDL/D, for the two deepest Zephyr hiders and the Deep Ignorance set (the other Zephyr models in decisions.md 2026-10-20: original -1.66 ± 0.44, ELM -1.26 ± 0.43, SimNPO -3.7 ± 1.8 bits, elicit; NPO and GradDiff mixed, their seeds scattering by ± 8 bits on curves that sit at 350). Rotated: the same sweep on A with every answer moved to another item, same n, same seeds, so the difference is paired point by point. The knowledge-specific curve is read by the rule calibrated in Table 8: a hump that survives the subtraction is teaching; a positive early value ($n \le 32$ here: the mean over three sizes and three seeds, beyond two seed standard errors) that crosses below zero later is a gate; a negative early value is elicitation; mixed within two standard errors. With three seeds that margin is coarse: CB and filter+CB+LAT sit at 2.0 and 1.9 standard errors and read mixed. The rule is kept as calibrated on the toy arms rather than loosened to these points; what it leaves is still the gate's pattern (every present-reading model at or below zero from $n = 8$ or 16; the three safeguards positive over $n \le 32$ on eight of nine seeds, crossing at 128–573), and the M21-n row makes the same separation without a margin. Its slope alone is not read, because every negative curve rises toward zero and the slope signature would call the original taught. M21 on B at the kept weights of each point, the same four rotations as Table 6; filter+CB+LAT reaches 1.19 at $n = 64$ and 1.39 at 128. The TinyStories model that cannot know the facts reads knowledge-specific −0.1 bits at $n = 573$ and M21 0.22.

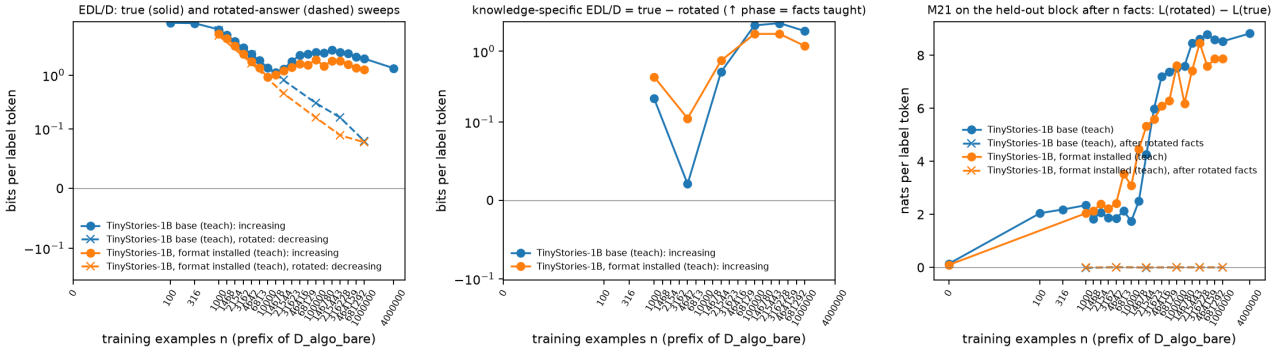
	taught		elicited		
	TinyStories base	TinyStories, format installed	op-bridge parent (same base)	Llama-3.2-1B base	Llama, format installed
EDL/D, true answers (bits): first, minimum, peak, last	9.4, 1.1, **2.9**, 1.3	5.9, 0.9, **1.9**, 1.3	6.5 → 0.13	3.5 → 0.04	0.97 → 0.04
EDL/D, answers rotated	6.8 → 0.08	5.5 → 0.08	4.7 → 0.08	2.3 → 0.33	0.93 → 0.07
knowledge-specific, true − rotated: n=1,000; peak (n); n=10^6	+0.2; +**2.5** (316K); +1.9	+0.4; +**1.8** (316K); +1.2	+0.4; +0.5 (100K); +0.06	+**1.2**; (first); −0.29	+0.04; +0.17 (21K); −0.03
read (M19k rule)	teach	teach	gate	gate, ≤ 0 from 21K	none
M21 on the held-out block: untrained → n=1,000 → 10^6 (nats/token)	0.13 → 2.3 → 8.5	0.09 → 2.0 → 7.9	1.15 → 7.2 → 13.5	0.57 → 10.8 → 15.0	2.7 → 9.0 → 17.1
n at half of M21's rise	31,600	14,700	3,200	$\le$ 1,000	1,500
M21 after rotated training, any n	0.00	0.00	0.00	0.01	0.01

Table 8: The two reads on the arithmetic Figure-2 arms (scaffold-free natural-language add/sub, 19 sizes from 1,000 to 10^6; the rotated twin at six of them, same per-size schedule; floors are each run's own minimum validation loss on the true stopping block). The taught arms keep their hump in the knowledge-specific curve, the rotated twin having none; the elicit arms keep nothing above 0.5 bits; the Llama base pays its format at the first size. M21 on the 600 held-out problems after n training examples; its half-rise size separates the regimes by one to two orders of magnitude, and the gated WMDP models of Table 7 reach theirs by 16–64 facts. The M19k rule, on the early value (the mean over the first three sizes, with its standard error across seeds): teach if the curve rises from it by more than 0.5 bits to a maximum above tolerance that lies after the early block; else gate if the early value is positive beyond 0.1 bits and two standard errors (with the size from which the curve stays at or below zero); else elicit if negative by the same margin; none within 0.1 bits; mixed otherwise.

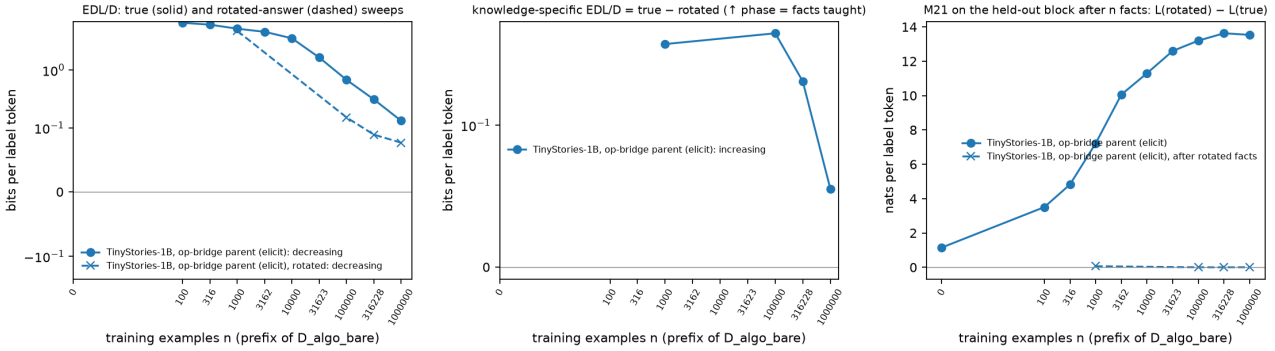


(a) Zephyr-7B: original and six unlearning methods

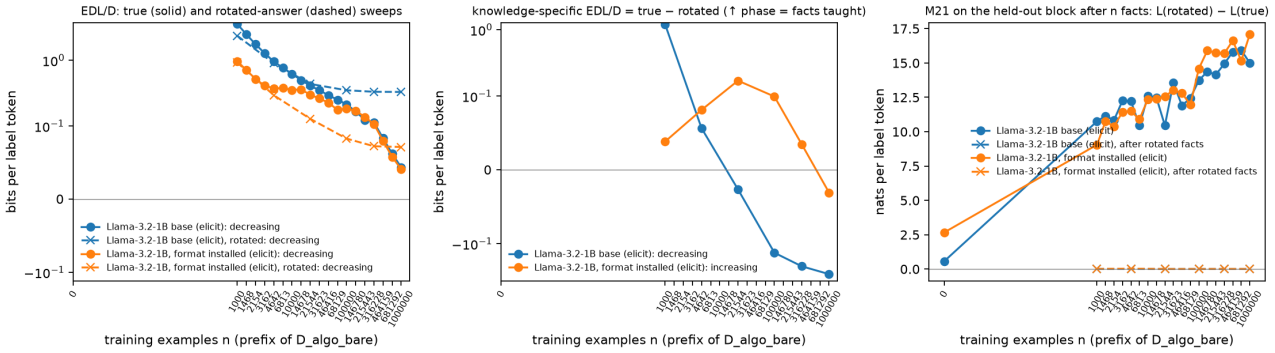
Figure 3: The two costs on WMDP-bio (Table 7). Left: EDL/D of the rotated-answer sweep, the format and unlock cost alone. Middle: the knowledge-specific curve, EDL/D(true) − EDL/D(rotated), seed mean (thick) and seeds (thin), symmetric log scale; below zero from the first sizes in every present-reading model, above zero and crossing later in the three safeguards. Right: M21 on the held-out B facts at the kept weights of every point, after true facts (solid) and after rotated facts (dashed). Own-minimum-validation floor throughout.



(a) taught: TinyStories-1B base and format-installed



(b) elicited: the op-bridge parent on the same base



(c) elicited: Llama-3.2-1B base and format-installed

Figure 4: The same two reads on the arithmetic Figure-2 arms (Table 8). Left: EDL/D of the true sweep (solid) and of the rotated-answer twin (dashed); the taught arms' hump is absent from the twin. Middle: the knowledge-specific curve, which keeps the hump on the taught arms (+2.5 and +1.8 bits at $n = 316{,}000$) and nothing above 0.5 bits on the elicited ones; the Llama base pays its format at $n = 1{,}000$ and crosses below zero by 21,000. Right: M21 on the 600 held-out problems after n examples, half of its rise by 15,000–32,000 examples on the taught arms and by $\leq 3{,}200$ on the elicited ones; 0.00 after rotated training.

Caveats

- There is no never-learned version of this model: "elicit" means "like the original", and the teach values come from each metric's own null and from the main results. The one cross-model anchor, the main results' TinyStories-1B twin on the same bio facts, is a 1B story model: it fixes a zero (0.05 nats on the rotated-answer twin, 0.22 on M21), not a scale, and a 7B model without bio knowledge does not exist: the Deep Ignorance strong-filter model, pretrained without the suite's biothreat-proxy data, still holds 0.92 of its baseline's M21 on these items (Section 5).
- The metrics before training are read on all bio items, those after relearning on half B; the first 512 items in source order score lower (original 0.51) than B (0.67), so the two blocks are not compared with each other.
- The multiple-choice circuit metrics (M1, M2, M4, the necessity of the original's heads, M1$^\bigstar$) read the shared answering machinery, as their MMLU controls show (Section 4), and are not tallied; the tallied circuit reading is the no-options one (M1f). The probe, the state swap and the option heads have MMLU controls only. The lens has no signal at the middle layers on the no-options surface even for the original (its first-token accuracy there is 0.12), so that read is dropped.
- The no-options surface is a harder read than the multiple-choice one (the original picks the correct answer among the four at 0.44 on B, against 0.67 with the options shown), so its recovery test has less room: for ELM, NPO and SimNPO the unrelated fine-tune already reaches the original there.
- One seed (three for M19) and one relearning recipe; the checkpoint with the lowest validation loss is used because held-out accuracy moves by ± 0.1 between evaluations.
- M19's rising phase is not by itself a teach call: the Deep Ignorance LAT models pay it with the answers rotated too (Section 5, Table 7). The signature is read here together with its fact-free twin and with M21 along the sweep; on the TOFU secondary design the elicit anchor itself reads "increasing" under both floors over $n \leq 200$, so that row is not called there.
- M19 covers $n \leq 573$, all of half A. In Figure 1a the teaching hump comes after an initial fall and at far larger n, so a hump beyond 573 facts is not excluded by the sweep alone; but the teach arm there ends far above the elicit arm (2.0 vs 0.04 bits at $n = 10^6$), while here the unlearned curves converge on the original's.
- M19's OCV floor (the owner's default) sits on the harder validation rows (the first 10% of A in source order), which is why the original's curve ends below zero; the paper's test floor (loss on B) lifts every curve by 0.37–0.41 bits per label token at $n = 573$ (the original ends at -0.15 instead of -0.56) and changes no call. One of the original's three seeds rises from $n = 8$ to 16.

Appendix: how each metric is judged

Every call is a position between the model's own null (0) and a reference measured in this design (1), $r = (v - \text{null})/(\text{ref} - \text{null})$, with elicit at $r \geq 0.5$; shifting or rescaling a statistic together with its null and reference changes nothing, and no row is called by a raw value (the rows that were, M3, M5, M6, M7, are reported only). The only absolute constants are detection gates, which decide whether there is anything to measure: 3 standard errors above the null (probe, state swap, lens, recovery), permutation $p < 0.01$ (hidden preference; 2,000 draws), binomial $p < 0.01$ against 100 random head sets (necessity), an overlap at least 0.05 above type-matched chance (0.10 for split-half circuits), a performing map (mean logit difference above 1 nat on its own pairs), and, for the option heads, the full model following the swap on at least half the pairs. A gate that fails on the original reads "instrument fails" before the unlearned model is judged. These rules, the reference modes, the tally exclusions and the presence of a bio twin for every MMLU control are property-tested on synthetic statistics (`tests/experiments/scripts/test_verdict_wmdp_rules.py`); the EDL signature rule and the restore-best keeper have their own tests. Leakage checks: the halves are split by item with balanced

letters; the relearning validation rows (the first 10% of A) are disjoint from the training rows and B is never trained on; the fine-tuning null's MMLU items are excluded from the MMLU accuracy read; every model uses the original's tokenizer, so items, pairs and over-length drops are identical across models.

	metric	null, measured on the same model	elicit reference	called
before training				
M12	hidden preference	the options permuted	original	yes
M16′	answer probe	probe on shuffled labels	original	yes
M9	lens, middle layers	zero logit difference (±SE)	original	yes
M11	repeatable task circuit	type-matched chance	original	yes
M14	answer carried by the state	rate with no swap	original	yes
M15	option-reading heads	no heads found	original	yes
M12f	answer preferred, no options shown	the "correct" candidate reassigned	original	yes
M11f	repeatable circuit, no options shown	type-matched chance	original	yes
M1★	original's heads necessary in U	100 random head sets	original on itself	machinery
M10★	original's head states patched in	32 random heads	original (the donor)	yes
M2★	wiring, U vs original	chance	split-half ceiling	yes
M13	accelerating descent	none	none: the original sits at a minimum	no
M7★	state change, original to U	the parent's own states	describes the unlearning	no
M21	hidden preference, no options, full answer	the answers rotated across items	original; cannot-know model 0.22	reported
after relearning				
M1f	circuit overlap, no options shown	type-matched chance	split-half ceiling	yes
M17f	held-out recovery, no options shown	the fine-tuning null	original	yes
M1, M2, M4	circuit overlap with the original	type-matched chance	split-half ceiling	machinery
M1	original's heads necessary in the child	100 random head sets	original's relearned child	machinery
M10	child's head states patched into U	32 random heads	the child (the donor)	yes
M17	held-out recovery	the fine-tuning null	original	yes
M17u	an unrelated fine-tune unlocks B	U itself	original	yes
M18	relearning cost	the fine-tuning null	original's relearning	one-sided
M19	EDL sweep	the paper's shape rule	original's own curve	yes
M17s	what the true answers add on B	relearning on rotated answers	original's relearning; cannot-know model 0.05	reported
M3	head roles	chance	none measured here	no
M5	gradient fades	none	original's relearning, 3.2×	no
M6	weight write	none	original's relearning, 8.8×10^{-3}	no
M7	one shared push	none	original's relearning, 0.64	no

Table 9: What a single value is compared with. A metric is called only when its null and the original's value are both measured here; the rest are reported beside the original's value, since their thresholds come from the main results (full fine-tuning of a 1B model), whose scale does not carry over to LoRA on a 7B model. "Machinery": the MMLU control reads the same as the bio reading (Table 4), so the row is graded but not tallied. Every before-training metric above M13 also has its MMLU control; the probe, the state swap and the option heads have no no-options version, because their labels (option index, letter swap) need the options.